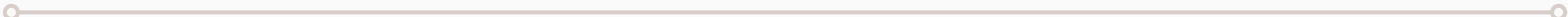




Volatility of Treasury Yields During Election Cycles



Sonakshi Jain, Kenny Nguyen, Khadija Oussabban, Alexander Tuosto, Virginia
Vaquero, Lucas Zenobio

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01

Introduction & Background on Treasury Markets

A brief overview of treasury securities, their purpose, and their role of financial markets.

What Are U.S. Treasury Securities?

- U.S. government debt instruments used to raise capital; backed by the full faith and credit of the U.S. government.
- Considered risk-free assets in global finance.
- Three main maturities:
 - Bills — under 1 year
 - Notes — 2 to 10 years
 - Bonds — 20 to 30 years
- Treasury yields serve as benchmark rates for mortgages, corporate bonds, and valuation models.

Why Treasuries Matter to Financial Markets

- Core financing tool for U.S. government spending and deficits.
- Define the risk-free rate used in pricing and discounting across finance.
- Anchor interest rates throughout the economy (loans, mortgages, corporate debt).
- Highly liquid → allows efficient trading, hedging, and price discovery.
- Integral to monetary policy (Fed open market operations).

Who Buys Treasuries & Why Investors Hold Them

Who Buys Them

- Institutional investors (pensions, insurers, asset managers)
- Banks (liquidity + collateral)
- Foreign governments/central banks (reserves)
- Federal Reserve (policy operations)
- Retail investors (TreasuryDirect, ETFs, funds)

Why Investors Hold Them

- Safe, low-default-risk asset with predictable cash flows.
- Highly liquid and easy to trade.
- Diversifies portfolios and hedges against risk assets.
- Hold value during recessions & uncertainty (“flight-to-quality”).

Yield Differences Across Tenors & Why We Study Volatility

Yield Differences Across Tenors

- Yields vary by maturity due to interest rate expectations.
- Longer maturities carry more inflation and interest rate risk.
- Yield curve shape (normal, flat, inverted) reflects economic outlook.

Why We Study Treasury Volatility

- Treasuries react quickly to economic and political uncertainty.
- Election cycles influence expectations around policy and inflation.
- Volatility helps gauge sentiment, risk pricing, and the stability of financial markets.

02

Data Sources & Variable Construction

A summary of the datasets, time period, and variables used to measure yield volatility.



What Question are we trying to answer?

Does the presence of a U.S. presidential election year affect volatility in Treasury yields?

We test this across three maturities:

3-Month

1-Year

10-Year

Where Did The Data Come From?

FRED (Federal Reserve)

- 10-Year Treasury Yield / DGS10 (Daily)
- 1-Year Treasury Yield /DGS1 (Daily)
- 3-Month Treasury Yield /DGS3MO (Daily, available since 1982)
- Recession Indicator /USREC (Daily NBER recession dummy)
- USD/EUR Exchange Rate /DEXUSEU (Daily)
- Britannica: Historical U.S. Presidential Election Dates

Why We Use Each Data Series?

- Treasury Yields (10Y, 1Y, 3M): Measure interest-rate movements and compute daily returns → core of volatility analysis.
- Recession Dummy (USREC): Controls for macroeconomic stress, ensures volatility results aren't "fake election effects."
- Exchange Rate (DEXUSEU): Controls for international risk sentiment → reduces omitted variable bias.
- Election Dates: Creates a binary election-year indicator to test whether elections change Treasury volatility.

What We Did With the Data

- Calculate Returns & Examine the Data
- R packages used: quantmod, rugarch, tseries, mfx, zoo
- Code steps performed:
 - Clean data (na.omit)
 - Calculate returns
 - Run GJR-GARCH models
 - Run probit regression
 - Perform t-tests

03

Methodology & Statistical Framework

An outline of the statistical tests and models applied to analyze Treasury return behavior.

Volatility Modeling Approach

- Applied rolling 30-day standard deviation to visualize short-term volatility across tenors.
- GJR-GARCH modeling captures conditional, time-varying volatility, also models asymmetry in response to negative returns (higher volatility after negative returns)
- Models estimated separately with and without election-year observations to quantify election-specific impact.
- Comparison across tenors demonstrates how short-, medium-, and long-term yields respond differently to shocks.

Statistical Tests & Predictive Modeling

- Two-sample t-tests compare mean volatility in election vs non-election years across tenors.
- Probit regression models probability of extreme volatility (over two standard deviations) as a function of election year dummy, recession dummy, exchange rate, and 10-year return.
- Approach controls for economic conditions while isolating the effect of presidential elections on volatility.
- Tests verify whether observed differences in volatility are statistically significant and economically meaningful.

04

Volatility Modeling & Empirical Results

A presentation of GJR-GARCH-based volatility estimates and key findings across tenors.

Summary Statistics: 3-Month

```
> basicStats(DGS3M0)
```

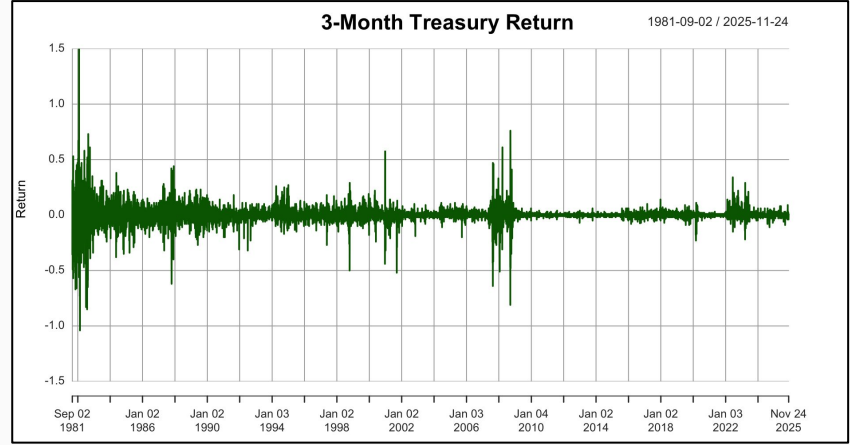
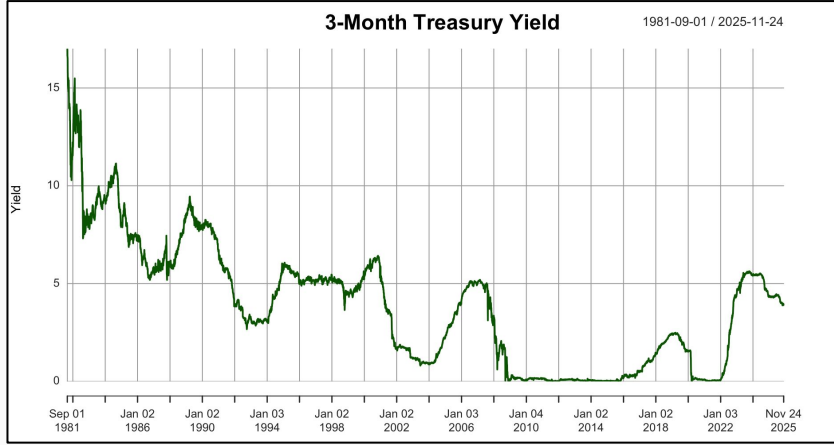
	DGS3M0
nobs	11057.000000
NAs	0.000000
Minimum	0.000000
Maximum	17.010000
1. Quartile	0.920000
3. Quartile	5.610000
Mean	3.829497
Median	4.000000
Sum	42342.750000
SE Mean	0.029841
LCL Mean	3.771003
UCL Mean	3.887991
Variance	9.846119
Stdev	3.137853
Skewness	0.639210
Kurtosis	0.154526

```
> basicStats(return_DGS3M0)
```

	DGS3M0
nobs	11056.000000
NAs	0.000000
Minimum	-1.040000
Maximum	1.690000
1. Quartile	-0.020000
3. Quartile	0.010000
Mean	-0.001185
Median	0.000000
Sum	-13.100000
SE Mean	0.000690
LCL Mean	-0.002537
UCL Mean	0.000167
Variance	0.005261
Stdev	0.072533
Skewness	-0.003943
Kurtosis	54.798724

3-Month Treasury Volatility

- Rolling volatility and GJR-GARCH forecasts show clear spikes during presidential election years and negative-return periods.
- Volatility peaks coincide with major election cycles, such as 2008, 2012, and 2016.
- Results highlight short-term sensitivity to political events and market uncertainty, consistent with expectations of increased risk during elections.



Augmented Dickey-Fuller Test

data: DGS3M0
Dickey-Fuller = -3.2433, Lag order = 22, p-value = 0.08028
alternative hypothesis: stationary

Augmented Dickey-Fuller Test

data: return_DGS3M0
Dickey-Fuller = -19.081, Lag order = 22, p-value = 0.01
alternative hypothesis: stationary

Warning message:

In `adf.test(return_DGS3M0)` : p-value smaller than printed p-value

Summary Statistics: 1-Year

```
> basicStats(DGS1)
```

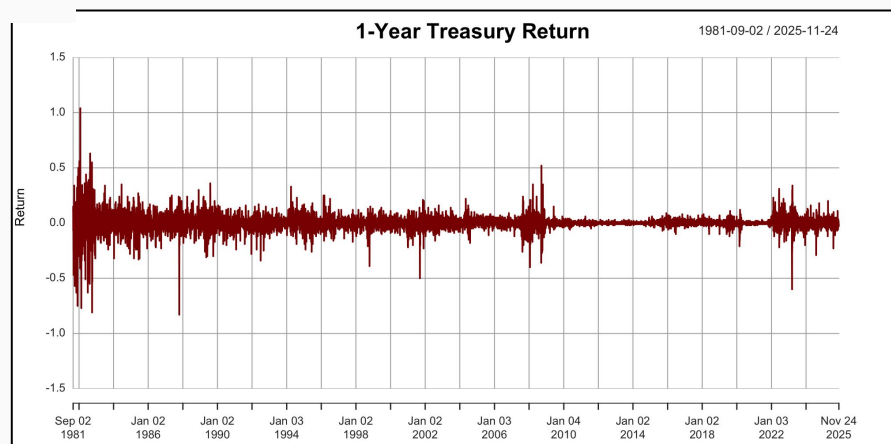
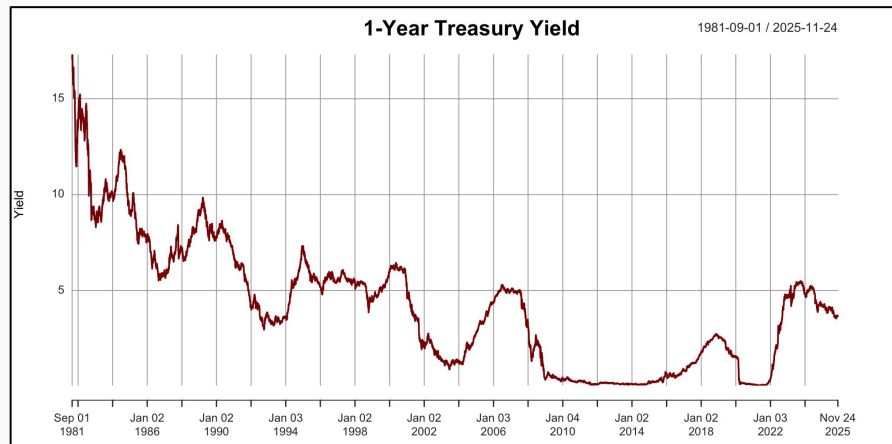
	DGS1
nobs	16675.000000
NAs	712.000000
Minimum	0.040000
Maximum	17.310000
1. Quartile	2.295000
3. Quartile	6.680000
Mean	4.885847
Median	4.920000
Sum	77992.770000
SE Mean	0.026625
LCL Mean	4.833658
UCL Mean	4.938035
Variance	11.316348
Stdev	3.363978
Skewness	0.678554
Kurtosis	0.632994

```
> basicStats(return_DGS1)
```

	DGS1
nobs	11057.000000
NAs	0.000000
Minimum	-0.830000
Maximum	1.040000
1. Quartile	-0.020000
3. Quartile	0.020000
Mean	-0.001216
Median	0.000000
Sum	-13.450000
SE Mean	0.000623
LCL Mean	-0.002438
UCL Mean	0.000005
Variance	0.004294
Stdev	0.065527
Skewness	-0.793292
Kurtosis	27.381406

1-Year Treasury Volatility

- Volatility patterns are smoother than 3-month yields with no consistent or large increase in volatility during election years.
- GJR-GARCH captures asymmetry: negative returns amplify volatility more than positive returns.
- Moderate volatility spikes suggest medium-term instruments partially absorb market uncertainty while still reacting to political events.



Augmented Dickey-Fuller Test

data: DGS1
Dickey-Fuller = -3.4674, Lag order = 22, p-value = 0.04532
alternative hypothesis: stationary

Augmented Dickey-Fuller Test

data: return_DGS1
Dickey-Fuller = -18.449, Lag order = 22, p-value = 0.01
alternative hypothesis: stationary

Warning message:

In `adf.test(return_DGS1)` : p-value smaller than printed p-value

10-Year Treasury Volatility

- Long-term yields remain stable, with no consistent volatility increase during election periods.
- GJR-GARCH results show asymmetry present but less pronounced than in short-term yields.
- Results confirm that institutional and long-term investors are less exposed to short-term political shocks, and election years do not materially change long-run volatility.

Summary Statistics: 10-Year

```
> basicStats(DGS10)
```

	DGS10
nobs	16675.000000
NAs	712.000000
Minimum	0.520000
Maximum	15.840000
1. Quartile	3.880000
3. Quartile	7.550000
Mean	5.822515
Median	5.460000
Sum	92944.800000
SE Mean	0.023290
LCL Mean	5.776864
UCL Mean	5.868165
Variance	8.658440
Stdev	2.942523
Skewness	0.737752
Kurtosis	0.363017

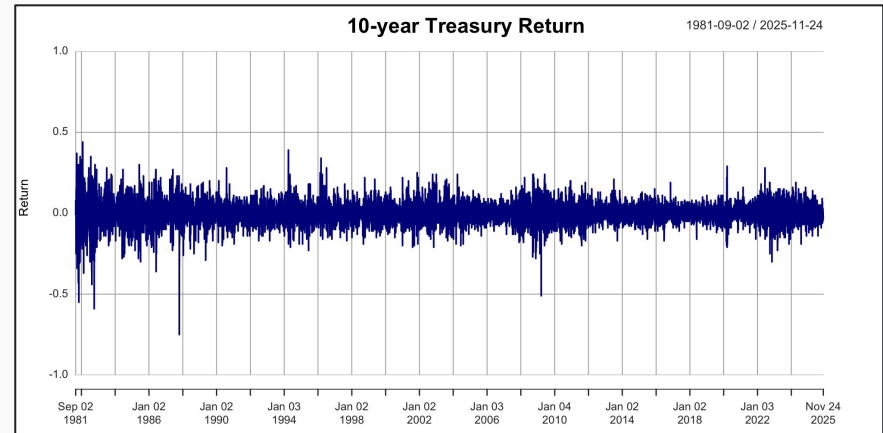
```
> basicStats(return_DGS10)
```

	DGS10
nobs	11057.000000
NAs	0.000000
Minimum	-0.750000
Maximum	0.440000
1. Quartile	-0.040000
3. Quartile	0.030000
Mean	-0.001028
Median	0.000000
Sum	-11.370000
SE Mean	0.000634
LCL Mean	-0.002272
UCL Mean	0.000215
Variance	0.004448
Stdev	0.066695
Skewness	-0.254684
Kurtosis	5.589078



Augmented Dickey-Fuller Test

```
data: DGS10
Dickey-Fuller = -3.5346, Lag order = 22, p-value = 0.03884
alternative hypothesis: stationary
```



Augmented Dickey-Fuller Test

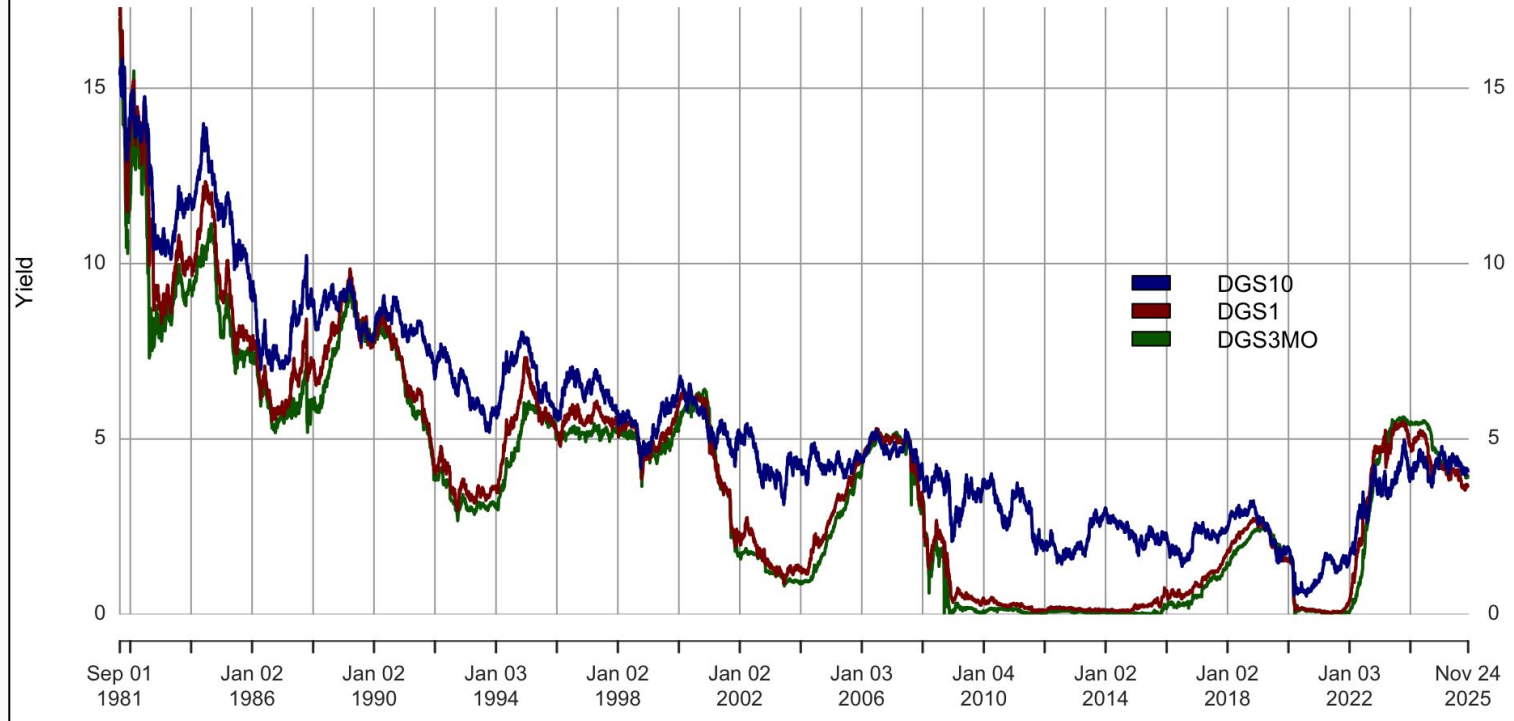
```
data: return_DGS10
Dickey-Fuller = -20.536, Lag order = 22, p-value = 0.01
alternative hypothesis: stationary
```

Warning message:

```
In adf.test(return_DGS10) : p-value smaller than printed p-value
```

Treasury Yield by Tenor

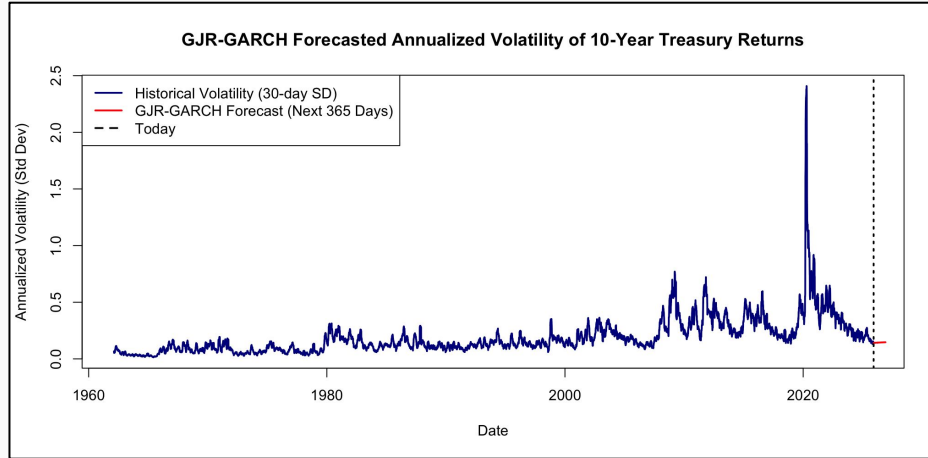
1981-09-01 / 2025-11-24



Parameter Estimates

- Parameters (alpha, beta, gamma) indicate persistence and asymmetry of volatility across tenors.
- 3M yields: highest sensitivity to shocks, strongest asymmetric response.
- 1Y yields: moderate sensitivity, noticeable asymmetric effects.
- 10Y yields: lowest sensitivity, indicating long-term market stability.
- Model outputs validate the choice of GJR-GARCH to capture real-world volatility patterns observed in the data.

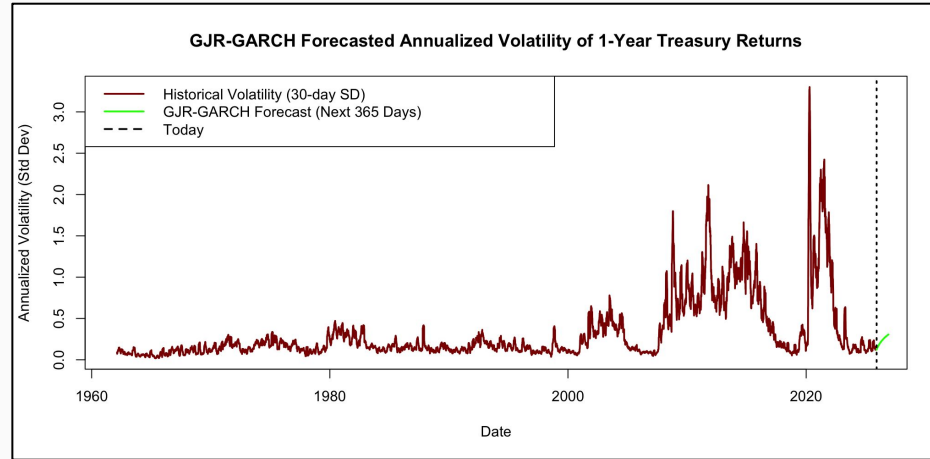
10-Year Return GJR GARCH estimate



T+360	0 0.009250
T+361	0 0.009251
T+362	0 0.009252
T+363	0 0.009253
T+364	0 0.009253
T+365	0 0.009254

Very low volatility predicted for the 10-year yield moving forward 365 days

1-Year Return GJR GARCH estimate



T+360	0 0.019204
T+361	0 0.019223
T+362	0 0.019241
T+363	0 0.019260
T+364	0 0.019278
T+365	0 0.019297

Less than 2% volatility for 1-year returns forecast,
higher than 10-year

3-Month GJR-GARCH estimate

- Extremely Fat tails, Kurtosis of 55
 - Often suffer from outliers
- Left out as outliers would make predictions invalid

05

Hypothesis Testing & Predictive Models

A review of t-tests and regression models evaluating how elections influence volatility.

Two-Sample t-Test Results

- 3M and 1Y yields: mean volatility appears higher, but not statistically significant in election years ($p > 0.05$).
- 10Y yields: smaller mean difference, not statistically significant
- Indicates short- and medium-term Treasuries are most impacted by presidential elections, supporting the hypothesis of election-induced volatility.

Two Sample t-test Results (10-Year)

```
Welch Two Sample t-test

data: more2vol by aligned$DGS10election
t = 1.5062, df = 161.09, p-value = 0.134
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -0.01242159  0.09226767
sample estimates:
mean in group 0 mean in group 1
  0.05844156      0.01851852
```

Different means, Not Statistically significant (p-value > 0.05, fail to reject null hypothesis of difference in means = 0)

Two Sample t-test Results (1-Year)

Welch Two Sample t-test

data: more2vol1 by aligned1\$DGS1election

t = 0.078558, df = 94.197, p-value = 0.9376

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0

95 percent confidence interval:

-0.07005485 0.07582685

sample estimates:

mean in group 0 mean in group 1

0.05844156 0.05555556

Different means, Not Statistically significant (p-value > 0.05, fail to reject null hypothesis of difference in means = 0)

Two Sample t-test Results (3-Month)

```
data: more2vol3 by aligned3$DGS3M0election
t = -1.2656, df = 72.146, p-value = 0.2097
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -0.15235060  0.03402448
sample estimates:
mean in group 0 mean in group 1
  0.05194805      0.11111111
```

Different means, Not Statistically significant ($p\text{-value} > 0.05$, fail to reject null hypothesis of difference in means = 0)

Probit Regression Results

- Election year dummy is not statistically significant: increases probability of extreme volatility (>2 SD) across tenors.
- Recession dummy significant: macroeconomic downturns amplify extreme volatility probability.
- Exchange rate and 10-year returns: smaller but relevant influences, demonstrating cross-market effects on Treasury volatility.
- Regression results complement GARCH findings, confirming both magnitude and probability of volatility spikes are driven more by macro conditions than elections.

10-Year Probit

```
Call:
glm(formula = more2vol ~ aligned$DGS10election + aligned$USREC +
    aligned$DEXUSEU, family = binomial(link = "probit"))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.6323	0.1711	-9.542	<2e-16 ***
aligned\$DGS10election	-0.7419	0.4973	-1.492	0.1357
aligned\$USREC	0.8098	0.4600	1.761	0.0783 .
aligned\$DEXUSEU	-1.9290	22.4356	-0.086	0.9315

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80.210 on 207 degrees of freedom
Residual deviance: 75.757 on 204 degrees of freedom
AIC: 83.757

Number of Fisher Scoring iterations: 7

Probit Coefficients (uninterpretable)

```
Call:
probitmfx(formula = more2vol ~ DGS10election + USREC + DEXUSEU,
    data = df, atmean = FALSE)
```

Marginal Effects:

	dF/dx	Std. Err.	z	P> z
DGS10election	-0.050490	0.024683	-2.0455	0.0408 *
USREC	0.127020	0.102738	1.2363	0.2163
DEXUSEU	-0.182516	2.123415	-0.0860	0.9315

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:

```
[1] "DGS10election" "USREC"
```

```
> |
```

Marginal effects:

On average, the presence of a Presidential Election decreases the probability that the 10-year treasury will experience >2 S.D. Volatility by ~ 5%

1-Year Probit

```
Call:
glm(formula = more2vol1 ~ aligned1$DGS1election + aligned1$USREC
    aligned1$DEXUSEU, family = binomial(link = "probit"))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.6051	0.1679	-9.560	<2e-16 ***
aligned1\$DGS1election	-0.1337	0.3461	-0.386	0.699
aligned1\$USREC	0.4735	0.4476	1.058	0.290
aligned1\$DEXUSEU	-22.9147	20.5647	-1.114	0.265

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 91.757 on 207 degrees of freedom
Residual deviance: 89.220 on 204 degrees of freedom
AIC: 97.22

Number of Fisher Scoring iterations: 6

Probit Coefficients (uninterpretable)

```
Call:
probitmfx(formula = more2vol1 ~ DGS1election + USREC + DEXUSEU,
    data = df1, atmean = FALSE)
```

Marginal Effects:

	dF/dx	Std. Err.	z	P> z
DGS1election	-0.014311	0.035177	-0.4068	0.6841
USREC	0.071400	0.085985	0.8304	0.4063
DEXUSEU	-2.581019	2.352663	-1.0971	0.2726

dF/dx is for discrete change for the following variables:

[1] "DGS1election" "USREC"

Marginal effects:

On average, the presence of a Presidential Election decreases the probability that the 1-year treasury will experience >2 S.D. Volatility by ~ 1%, NOT SIGNIFICANT

3-Month Probit

```
Call:
glm(formula = more2vol3 ~ aligned3$DGS3M0election + aligned3$USREC +
    aligned3$DEXUSEU, family = binomial(link = "probit"))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.6441	0.1745	-9.424	<2e-16 ***
aligned3\$DGS3M0election	0.4764	0.2947	1.617	0.106
aligned3\$USREC	-0.2633	0.5507	-0.478	0.633
aligned3\$DEXUSEU	-26.9011	20.6264	-1.304	0.192

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 102.593 on 207 degrees of freedom
Residual deviance: 98.889 on 204 degrees of freedom
AIC: 106.89

Number of Fisher Scoring iterations: 6

Probit Coefficients (uninterpretable)

```
Call:
probitmfx(formula = more2vol3 ~ DGS3M0election + USREC + DEXUSEU,
    data = df3, atmean = FALSE)
```

Marginal Effects:

	dF/dx	Std. Err.	z	P> z
DGS3M0election	0.069768	0.049400	1.4123	0.1579
USREC	-0.028160	0.049433	-0.5697	0.5689
DEXUSEU	-3.384082	2.635331	-1.2841	0.1991

dF/dx is for discrete change for the following variables:

```
[1] "DGS3M0election" "USREC"
```

Marginal effects:

On average, the presence of a Presidential Election increases the probability that the 3-Month treasury will experience >2 S.D. Volatility by ~ 7%, NOT SIGNIFICANT

06

Conclusions, Implications & Future Extensions

A discussion of major takeaways, investor impacts, and ideas for expanding the project.

Key Conclusions

- Election years do not consistently increase volatility in Treasury yields across maturities.
 - 10-Year: stable or slightly lower volatility
 - 1-Year: no meaningful difference
 - 3-Month: higher raw volatility, but not statistically significant
- Macro forces (recessions, USD movements) drive Treasury volatility more than election cycles.
- T-tests and probit all point to limited election impact relative to broader economic conditions.

Investor Implications

Retail Investors

- Do not overreact to election headlines or campaign noise.
- Treasuries remain a safe, stable asset, even in political uncertainty.
- Election timing is not a reliable trading strategy, focus instead on long-term asset allocation.

Institutional Investors

- Election years do not justify major Treasury risk shifts.
- Only slight monitoring needed for short-end (3M) volatility.
- GJR-GARCH volatility models remain useful, but macro data dominates political variables for rate forecasts.

Future Extensions

- Add more variables: Fed rate cuts, candidate party affiliation, macro indicators, expected future short-term rates.
- Refine time windows: Compare pre- vs. post-election periods to isolate anticipatory market behavior.
- Expand models:
 - Intraday volatility analysis
 - Multivariate GARCH to examine cross-tenor volatility spillovers
 - Global yield-curve comparisons
- These extensions increase predictive power and improve practical guidance during political and economic uncertainty.

References

- Anil K Kashyap, Jeremy C. Stein, et al. “What’s Going on in the US Treasury Market, and Why Does It Matter?” *Brookings*, 15 Apr. 2025, www.brookings.edu/articles/whats-going-on-in-the-us-treasury-market-and-why-does-it-matter/.
- Moench, Emanuel, and Soroosh Soofi-Siavash. “What Moves Treasury Yields?” *Journal of Financial Economics*, vol. 146, no. 3, 2022, pp. 1016–1043. Elsevier, <https://doi.org/10.1016/j.jfineco.2022.04.001>
- *Understanding the Importance of Measuring U.S. Treasury Volatility Q3 2025*, www.cmegroup.com/articles/files/2025/understanding-the-importance-of-measuring-us-treasury-volatility.pdf. Accessed 3 Dec. 2025.



Thank You!

